Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Calculate ΔG^{o} for the following cell reaction
$Zns + Ag_2Os + H_2Ol \rightarrow Zn^{2+}aq + 2Ags + 2OH^{-}aq$

$E^{0}_{Ag^{+}/Ag} = +0.80V$ and $E^{0}_{Zn^{+2}/Zn} = -0.76V$,F=96500

(Given: $K_{sp}ofAgOH = 2 \times 10^{-8}$)

(A) $-305kJ/mol$

(B) $212\ kJ/mol$

(C) $305kJ/mol$

(D) $301kJ/mol$

Q.2

For an electrolyte solution of $0.05\ mol\ L^{-1}$, the conductivity has been found to be $0.0110\ S\ cm^{-1}$. The molar conductivity is

(A) $0.055\ S\ cm^2\ mol^{-1}$

(B) $550Scm^2\ mol^{-1}$

(C) $0.22\ S\ cm^2\ mol^{-1}$

(D) $220\ S\ cm^2\ mol^{-1}$

Q.3

The highest electrical conductivity of the following aqueous solutions is of

(A) $0.1M$ difluoroacetic acid

(B) $0.1M$ fluoroacetic acid

(C) $0.1M$ chloroacetic acid

(D) $0.1M$ acetic acid

Q.4

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The specific conductivity of $N/10KCl$ solution at $20^{\circ}C$ is $0.212 ohm^{-1}\ cm^{-1}$ and the resistance of the cell containing this solution at $20^{\circ}C$ is $55 ohm$. The cell constant is

(A) $4.616\ cm^{-1}$

(B) $11.66\ cm^{-1}$

(C) $2.173\ cm^{-1}$

(D) $3.324\ cm^{-1}$

Q.5

The equivalent conductance at infinite dilution of a weak acid such as HF

(A) can be determined by extrapolation of measurements of dilute solutions of HCl, HBr and HI

(B) can be determined by measurement of very dilute HF solutions

(C) can be determined from measurements of dilute solutions of NaF, $NaCl$ and HCl

(D) is an undefined quantity

Q.6

The standard reduction potential for Cu^{2+}/Cu is $+\ 0.34\ V$. Calculate the reduction potential at $pH = 14$ for the above couple. K_{sp} of $CuOH_2 = 1 \times 10^{-19}$

(A) $-0.22\ V$

(B) $+0.22V$

(C) $-0.44V$

(D) $+0.44V$

Q.7

Given the ionic conductance of

$$\begin{array}{l} COO^- \\ | \\ COO^- \end{array}$$

Na^+ and K^+ are 74,50 , and $73\ cm^2 ohm^{-1} eq^{-1}$, respectively. The equivalent conductance at infinite dilution of the salt

$$\begin{array}{l} COONa \\ | \\ COOK \end{array}$$

is

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(A) $197\ \mathrm{cm^2 ohm^{-1} eq^{-1}}$

(B) $172\ \mathrm{cm^2 ohm^{-1} eq^{-1}}$

(C) $135.5\ \mathrm{cm^2 ohm^{-1} eq^{-1}}$

(D) $160.5\ \mathrm{cm^2 ohm^{-1} eq^{-1}}$

Q.8

When electric current is passed through a cell having an electrolytic solution, the cations move towards the cathode and anions towards the anode. If anode is pulled out from the solution

(A) the cations and anions will move towards the cathode

(B) the cations will continue to move towards cathode and anions will stop moving

(C) both the cations and anions will stop moving

(D) the cations and anions will start moving randomly.

Q.9

EMF of the following cell is 0.6 volt.

$\mathrm{AgsAgBrsKBr0.01M} \mid\ \mid \mathrm{AgNO_3 0.001\ M} \mid \mathrm{Ag\ (s)}$

K_{sp} of AgBr is expressed as 1×10^{-x}, x is

Take $\frac{2.303RT}{F} = 0.06\ \ V$

(A) 13.00

(B) 15.00

(C) 14.00

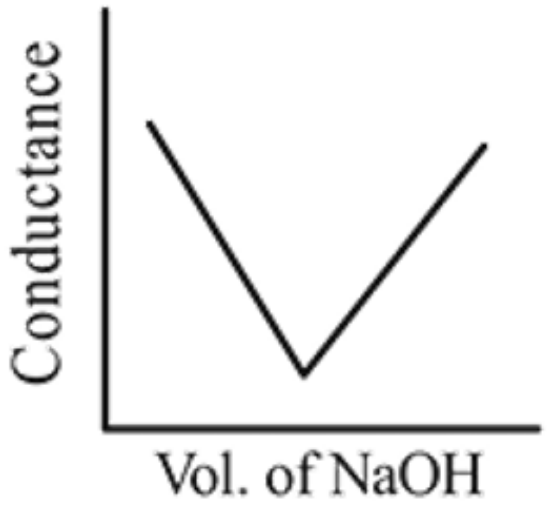
(D) 16.00

Q.10

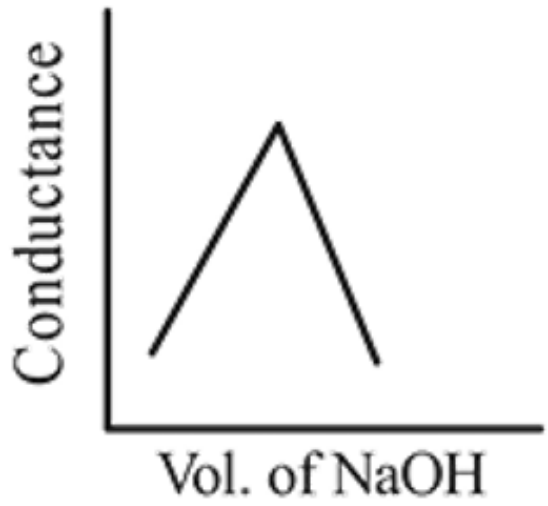
HNO_3 (aq) is titrated with $NaOH(aq)$
conductometrically, graphical representation of the titration is :

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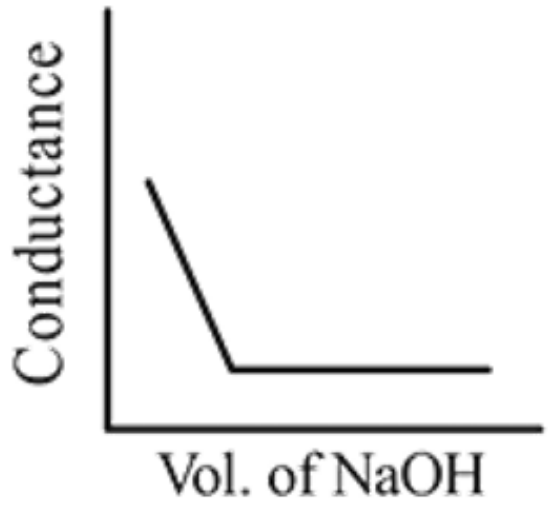
(A)



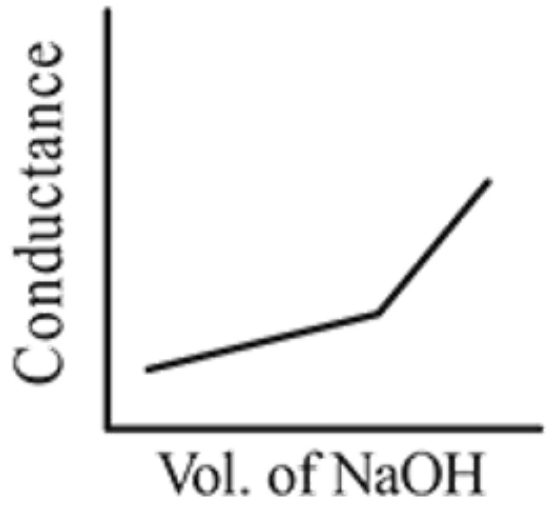
(B)



(C)



(D)



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Q.11

Resistance of a conductivity cell filled with a solution of an electrolyte of concentration $0.1\mathrm{M}$ is 100Ω. The conductivity of this solution is $1.29\ \mathrm{Sm}^{-1}$. Resistance of the same cell when filled with $0.02\mathrm{M}$ of the same solution is 520Ω. The molar conductivity of $0.02\mathrm{M}$ solution of electrolyte will be

(A) $1.24 \times 10^{-4}\ \mathrm{S\ m^2\ mol^{-1}}$

(B) $12.4 \times 10^{-4}\ \mathrm{S\ m^2\ mol^{-1}}$

(C) $124 \times 10^{-4}\ \mathrm{S\ m^2\ mol^{-1}}$

(D) $1240 \times 10^{-4}\ \mathrm{S\ m^2\ mol^{-1}}$

Q.12

$\Lambda_{\mathrm{ClCH_2COONa}} = 224\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{geq}^{-1}$
$\Lambda_{\mathrm{NaCl}} = 38.2\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{geq}^{-1}$
$\Lambda_{\mathrm{HCl}} = 203\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{geq}^{-1}$
What is the value of $\Lambda_{\mathrm{ClCH_2COOH}}$

(A) $288.5\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{geq}^{-1}$

(B) $289.5\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{geq}^{-1}$

(C) $388.8\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{geq}^{-1}$

(D) $59.5\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{geq}^{-1}$

Q.13

Specific conductance of $0.1\mathrm{M}$ sodium chloride solution is $1.06 \times 10^{-2}\mathrm{ohm}^{-1}\ \mathrm{cm}^{-1}$. Its molar conductance in $\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{mol}^{-1}$ is

(A) 1.06×10^2

(B) 1.06×10^3

(C) 1.06×10^4

(D) 5.3×10^2

Q.14

The resistance of $1\ \mathrm{N}$ solution of acetic acic is $250\mathrm{ohm}$, when measured in a cell of cell constant $1.15\ \mathrm{cm}^{-1}$. The equivalent conductance (in $\mathrm{ohm}^{-1}\ \mathrm{cm}^2$ equiv $^{-1}$) of $1\ \mathrm{N}$ acetic acid will be

(A) 4.6

(B) 9.2

(C) 18.4

(D) 0.023

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Q.15

Specific conductance of $0.1M\mathrm{HA}$ is $3.75 \times 10^{-4}\ \mathrm{ohm}^{-1}\ \mathrm{cm}^{-1}$. If $\lambda^{\infty}(\mathrm{HA}) = 250\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{mol}^{-1}$,
the dissociation constant K_a of HA is:

(A) 1.0×10^{-5}

(B) 2.25×10^{-4}

(C) 2.25×10^{-5}

(D) 2.25×10^{-13}

Q.16

At $25^{\circ}\mathrm{C}$, the resistance of a cell filled with $0.01\mathrm{MKCl}$ solution is 525 ohms. The resistance of the same cell filled with $0.1\mathrm{MNH_4OH}$ is 2030 ohms. By calculating the degree of dissociation, calculate the equilibrium constant of $\mathrm{NH_4OH}$.
Given : $\lambda^{\infty}\left(\mathbf{K}^{+}\right) = 73.52, \lambda^{\infty}\left(\mathbf{Cl}^{-}\right) = 76.34, \lambda^{\infty}\left(\mathbf{NH}_4^{+}\right) = 73.4$ and $\lambda^{\infty}\left(\mathrm{OH}^{-}\right) = 197.6\Omega^{-1}\ \mathrm{cm}^2\ \mathrm{mol}^{-1}$

(A) $2.05 \times 10^{-4}\mathrm{M}$

(B) $2.15 \times 10^{-5}\mathrm{M}$

(C) $1.05 \times 10^{-7}\mathrm{M}$

(D) $2.05 \times 10^{-5}\mathrm{M}$

Q.17

Resistance of a conductivity cell filled with a solution of an electrolyte of concentration $0.$ M is 100Ω. The condnctivity of this solution is $1.29\ \mathrm{S\ m}^{-1}$. Resistance of the same cell when filled with $0.2\mathrm{M}$ of the same solution is 520Ω. The molar conductivity of $0.02\mathrm{M}$ solution of the electrolyte will be

(A) $12.4 \times 10^{-4}\mathrm{Sm^2moT^{-1}}$

(B) $124 \times 10^{-4}\ \mathrm{S\ m^2mo^{-1}}$

(C) $1240 \times 10^{-4}\ \mathrm{S\ m^2\ mol^{-1}}$

(D) None of these

Q.18

Identify the correct statement :

(A) Corrosion of iron can be minimized by forming a contact with another metal with a higher reduction potential

(B) Iron corrodes in oxygen free water

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(C) Corrosion of iron can be minimized by forming an impermeable barrier at its surface

(D) Iron corrodes more rapidly in salt water because its electrochemical potential is higher

Q.19

Which of the following is a highly corrosive salt?

(A) Hg_2Cl_2

(B) $HgCl_2$

(C) $FeCl_2$

(D) $PbCl_2^2$

Q.20

The corrosion of iron is essentially an electrochemical phenomenon. What are the cell reactions involving in the phenomenon?

(A) Fe is oxidised to Fe^{3+} and H_2O is reduced to O_2^{2-}.

(B) Fe is oxidised to Fe^{2+} and H_2O is reduced to O_2^{-}.

(C) Fe is oxidised to Fe^{2+} and H_2O is reduced to O_2.

(D) Fe is oxidised to Fe^{2+} and dissolved oxygen in water is reduced to OH^{-}

Answers & Solutions

Q.1 Answer:

212 kJ / mol

Solution:

Calculation for $E^{-}_{OH^{-}Ag_2OAg}$

Reduction $[\ 2Ag^{+} + 2e^{-} \rightarrow 2Ags\ ,\ \Delta\ G_1^o = -2 \times F \times E^o{}_{Ag^{+}\ |\ Ag}$

Ionisation $[\ Ag_2O + H_2Ol \rightarrow 2Ag^{+} + 2OH^{-}\ \Delta\ G_2^o = -2.303RTlogk_{sp^2}$

Net $Ag_2O + H_2Ol + 2e^{-} \rightarrow 2Ags + 2OH^{-}\ ,\ \Delta\ G_3^o = -2 \times F \times E^o_{OH^{-}Ag_2OAg}$

$\Delta\ G_3^o = \Delta\ G_1^o + \Delta\ G_2^o$

$+ 2 \times F \times E^o_{OH^{-}Ag_2OAg} = +2 \times F \times E^o{}_{Ag^{+}\ |\ Ag} + \frac{2.303RTlogk_{sp^2}}{2F}$

$E^o_{OH^{-}Ag_2OAg} = E^o{}_{Ag^{+}\ |\ Ag} + \frac{0.06}{2} \times 2log\ k_{sp}$

$= 0.8 + 0.06log\ 2 \times 10^{-8}$

$= 0.8 + 0.06 - 8 + 0.3$

$= 0.8 + 0.06 \times -7.7$

$= 0.8 - 0.462$

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$= 0.338\text{Volt} \cong 0.34\text{V}\ \ \Delta\ G^{o} = \ -\ nFE^{o}_{ceu}$

$= \ -\ nFE^{o}_{c}\ -\ E_{A_{Red}}$

$= \ -\ 2 \times E^{o}_{OH^{-}Ag_{2}OAg}\ -\ E^{o}_{Zn^{+2}\ |\ Zn}$

$= \ -\ 2 \times 96500 \times 0.34 + 0.76$

$= \ -\ 2 \times 96500 \times 1.1\text{volt}$

$= 212300\text{J}$

$= 212.3\ \text{kJ}$

Q.2 Answer:

$220\ \text{S cm}^{2}\ \text{mol}^{-1}$

Solution:

$\Lambda_{m} = \frac{\kappa \times 1000}{M} = \frac{0.0110 \times 1000}{0.05}$
$= 220\ \text{S cm}^{2}\ \text{mol}^{-1}$

Q.3 Answer:

0.1M difluoroacetic acid

Solution:

Difluoroacetic acid being strongest acid will furnish maximum number of ions showing highest electrical conductivity.

Q.4 Answer:

$11.66\ \text{cm}^{-1}$

Solution:

Cell constant
$= \kappa \times \text{R}$
$= 0.212\text{ohm}^{-1}\ \text{cm}^{-1} \times 55\text{ohm}$
$= 11.66\ \text{cm}^{-1}$

Q.5 Answer:

can be determined from measurements of dilute solutions of NaF, NaCl and HCl

Solution:

Kohlrausch Law, $\Lambda_{eq}(\text{NaF}) + \Lambda_{eq}(\text{HCl}) -$
$\Lambda_{eq}(\text{NaCl}) = \Lambda_{eq}(\text{HF})$

Q.6 Answer:

$-\ 0.22\ \text{V}$

Solution:

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When $pH = 14,\ [H^+] = 10^{-14}$ and $[OH^-] = 1\ M$

$K_{sp} = [Cu^{2+}][OH^-]^2 = 10^{-19}$

$\therefore\ [Cu^{2+}] = \frac{10^{-19}}{[OH^-]^2} = 10^{-19}$

The half cell reaction

$Cu^{2+} + 2e^- \rightarrow Cu$

$E = E^0 - \frac{0.059}{2}\log\frac{1}{[Cu^{2+}]}$

$= 0.34 - \frac{0.059}{2}\log\frac{1}{10^{-19}} = -0.22\ V$

Q.7 Answer:

$135.5\ cm^2 ohm^{-1} eq^{-1}$

Solution:

Total charge $= 2$ **Number of equivalent of ion**

$= \frac{\text{Charge on the ion}}{\text{Total charge}}$

$\therefore$ **Eq of** $\begin{pmatrix} COO^- \\ COO^- \end{pmatrix} = \frac{2}{2} = 1$

Eq of $Na^+ = \frac{1}{2}$, **Eq of** $K^+ = \frac{1}{2}$

$\therefore \lambda^\circ_{eq}\begin{pmatrix} COONa \\ COOK \end{pmatrix}$

$= \lambda^\circ_{eq}\begin{pmatrix} COO^- \\ COO^- \end{pmatrix} + \frac{1}{2}\lambda^\circ Na^+ + \frac{1}{2}\lambda^\circ K^+$

$= 74 + \frac{50}{2} + \frac{73}{2} = 135.5 ohm^{-1}\ cm^2 eq^{-1}$

Q.8 Answer:

the cations and anions will start moving randomly.

Solution:

On pulling out one of the electrodes the electric field applied to the solution disappears and hence the ions start moving randomly.

Q.9 Answer:

15.00

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Solution:

$Ag \rightarrow Ag^{+}L + e^{-}$

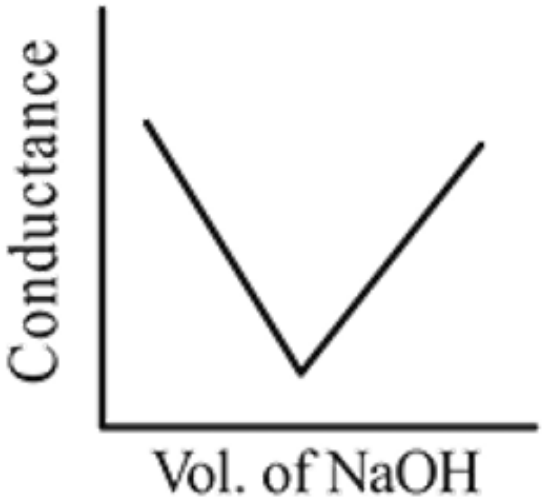
$Ag^{+}0.001\ M + e^{-} \rightarrow Ag$

$Ag^{+}\ (\ 0.001\ M\ )\ \rightarrow Ag^{+}\ (\ L\ )$

$0.6 = 0 - \frac{0.06}{1}\log\frac{K_{sp}}{10^{-2} \times 10^{-3}}$

$10^{-10} = \frac{K_{sp}}{10^{-5}} \Rightarrow K_{sp} = 10^{-15}$

Q.10 Answer:



Solution:

Molar conductivity of H^{+} and OH^{-} are very high as compare to other ions. Initially conductance of solution sharply decreases due to consumption of free H^{+}. After complete neutralization further slightly increases due to presence of OH^{-}.

Q.11 Answer:

$124 \times 10^{-4}\ S\ m^{2}\ mol^{-1}$

Solution:

Conductivity $(\kappa) = \frac{\ell}{R.a\ .}$ $\kappa = \frac{l}{R \cdot a}$**; cell constant.** $\left(\frac{l}{a}\right) = 1.29 \times 100 = 129$
Again conductivity of $0.02M$ **solution** $\kappa = \frac{1}{520} \times 129$
$\Lambda_{m} = \frac{\kappa \times 1000}{M} = \frac{129}{520} \times \frac{10^{-3}}{0.02}$
$= 124 \times 10^{-4}\ S\ m^{2}\ mol^{-1}$

Q.12 Answer:

$388.8 ohm^{-1}\ cm^{2}\ geq^{-1}$

Solution:

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$\mathrm{ClCH_2COONa} + \mathrm{HCl} \rightarrow \mathrm{ClCH_2COOH} + \mathrm{NaCl}$
$\Lambda_{\mathrm{ClCH_2COONa}} + \Lambda_{\mathrm{HCl}} = \Lambda_{\mathrm{ClCH_2COOH}} + \Lambda_{\mathrm{NaCl}}$
$224 + 203 = \Lambda_{\mathrm{ClCH_2COOH}} + 38.2$
$\Lambda_{\mathrm{ClCH_2COOH}} = 427 - 38.2$
$= 388.8 \mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{geq}^{-1}$

Q.13 Answer:

1.06×10^2

Solution:

$\Lambda_{\mathrm{m}} = \frac{\kappa \times 1000}{\mathrm{M}} = \frac{1.06 \times 10^{-2} \times 1000}{0.1}$
$= 1.06 \times 10^2$

Q.14 Answer:

4.6

Solution:

$$\kappa = \frac{\text{Cell constant}}{\mathrm{R}} = \frac{1.15}{250}$$
$$\Lambda_{\mathrm{eq}} = \frac{1.15}{250} \times \frac{1000}{1} = 4.6$$

Q.15 Answer:

2.25×10^{-5}

Solution:

$\lambda_m = \frac{1000\kappa}{0.1} = \frac{1000 \times 3.75 \times 10^{-4}}{0.1} = 3.75$
$\alpha = \frac{\lambda_{\mathrm{m}}}{\lambda_{\mathrm{m}}^{\infty}} = \frac{3.75}{250} = 1.5 \times 10^{-2}$
$K_{\mathrm{a}} = \mathrm{C}\alpha^2 = 0.1 \times \left(1.5 \times 10^{-2}\right)^2 = 2.25 \times 10^{-5}$

Q.16 Answer:

$2.05 \times 10^{-5} \mathrm{M}$

Solution:

Molar conductivity at infintite dilution of KCl is

$\Lambda_{\mathrm{m}}^{\infty}(\mathrm{KC}) = \lambda_{\mathrm{m}}^{\infty}\left(\mathrm{K}^{+}\right) + \lambda_{\mathrm{m}}^{\infty}\left(\mathrm{C}^{-}\right)$
$= (73.52 + 76.34) \Omega^{-1}\ \mathrm{cm}^2\ \mathrm{mol}^{-1} = 149.86 \Omega^{-} \mathrm{cm}^2\ \mathrm{mol}^{-1}$

Conductivity of $0.01 \mathrm{MKCl}$ is

$\mathbf{K}_1 = \Lambda_{\mathrm{m}}^{\infty} \mathrm{c} = \left(149.86 \Omega^{-1}\ \mathrm{cm}^2\ \mathrm{mol}^{-1}\right)$
$\left(0.01 \times 10^{-3}\ \mathrm{mol\ cm}^{-3}\right) = 1.4986 \times 10^{-3} \Omega^{-1}\ \mathrm{cm}^{-1}$

Cell constant is

$K = k_1 R = \left(1.4986 \times 10^{-3} \Omega^{-1}\ \mathrm{cm}^{-1}\right)(525 \Omega) = 0.7868\ \mathrm{cm}^{-1}$

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Conductivity of $0.1 \mathrm{M} \mathrm{NH}_4\mathrm{OH}$ is

$K_2=\frac{K}{R}=\frac{0.7868 \mathrm{~cm}^{-1}}{2030 \Omega}=3.876 \times 10^{-4} \Omega^{-1} \mathrm{~cm}^{-1}$ **Molar conductivity of** $0.1 \mathrm{M} \mathrm{NH}_4\mathrm{OH}$ **is**

$\Lambda_{\mathrm{o}}\left(\mathrm{NH}_4\mathrm{OH}\right)=\frac{k_2}{\mathrm{c}}=\frac{3.876 \times 10^{-1} \Omega^{-1} \mathrm{~cm}^{-1}}{0.1 \times 10^{-3} \mathrm{~mol} \mathrm{~cm}^{-3}}$

$=3.876 \Omega^{-1} \mathrm{~cm}^2 \mathrm{~mol}^{-1}$

Molar conducitivity of $\mathrm{NH}_4\mathrm{OH}$ **at infinite dilution is**

$\Lambda_{\mathrm{m}}^{\infty}\left(\mathrm{NH}_4\mathrm{OH}\right)=\lambda_{\mathrm{m}}^{\infty}\left(\mathrm{NH}_4^{+}\right)+\lambda_{\mathrm{m}}^{\infty}(\mathrm{OH})$

$=(73.4+197.6) \Omega^{-1} \mathrm{~cm}^2 \mathrm{~mol}^{-1}$

$=271.0 \Omega^{-1} \mathrm{~cm}^2 \mathrm{~mol}^{-1}$

Degree of dissociation of $\mathrm{NH}_4\mathrm{OH}$ **is**

$\alpha=\frac{\Lambda_{\mathrm{o}}}{\Lambda_{\mathrm{m}}^{\mathrm{o}}}=\frac{3.876 \Omega^{-1} \mathrm{~cm}^2 \mathrm{~mol}^{-1}}{271.0 \Omega^{-1} \mathrm{~cm}^2 \mathrm{~mol}^{-1}}=0.0143$

Equilibrium constant of $\mathrm{NH}_4\mathrm{OH}$ **is** $\mathrm{K}=\frac{\left[\mathrm{NH}_4^{+}\right]\left[\mathrm{OH}^{-}\right]}{\left[\mathrm{NH}_4\mathrm{OH}\right]}=\frac{(\mathrm{c} \alpha)(\mathrm{co})}{\mathrm{c}(1-\alpha)} \simeq \mathrm{ca}^2 \quad=$

$(0.1 \mathrm{M})(0.0143)^2=2.05 \times 10^{-5} \mathrm{M}$

Q.17 Answer:

$124 \times 10^{-4} \mathrm{~S} \mathrm{~m}^2 \mathrm{mo}^{-1}$

Solution:

Cell constant $=$

$\frac{\text{conductivity(l)}}{\text{condnctance(I)}}=\frac{\text{conductivity (II)}}{\text{condutarce(II)}}$ **Conductivity** $(\Pi)=\frac{129}{520} \sin^{-1}$

$0.2 \mathrm{M}=200 \mathrm{~mol} \mathrm{~m}^{-3}$

Molar conductivity of $0.2 \mathrm{M}$ **solution**

$=\frac{129 / 520}{200}=124 \times 10^{-4} \mathrm{sm}^2 \mathrm{md}^{-1}$

From the given data it is not possible to calculate molar condnctivity of $0.02 \mathrm{M}$ **solution exactly . Havever, it is sure that its molar conductivity will be slightly greater than** $12.4 \times 10^{-4} \mathrm{sm}^2 \mathrm{~mol}^{-1}$**. Assuming linear inverse variation of molar conductivity with concentration we can say that**

$\therefore$ **Molar conductivity of** $0.02 \mathrm{M}$ **solution**

$=1.24 \times 10^{-3} \times 10$

$=124 \times 10^{-4} \mathrm{~s} \mathrm{~m}^2 \mathrm{mo}^{-1}$

Q.18 Answer:

Corrosion of iron can be minimized by forming an impermeable barrier at its surface

Solution:

Corrosion of iron can be minimized by forming an impermeable barrier at its surface.

Q.19 Answer:

HgCl_2

Solution:

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$E^{\circ}_{Hg^{2+}|Hg} > E^{\circ}_{Hg_2^{2+}|Hg} > E^{\circ}_{Fe^{2+}|Fe} > E^{\circ}_{Pb^{2+}|Pb}$

Q.20 Answer:

Fe is oxidised to Fe^{2+} and dissolved oxygen in water is reduced to OH^-

Solution:

$2Fe \rightarrow 2Fe^{2+} + 4e^-$ **(anodic reaction)**

$O_2 + 2H_2O + 4e^- \longrightarrow 4OH^-$ **(cathodic reaction)**

The overall reaction is

$2Fe + O_2 + 2H_2O \rightarrow 2FeOH_2$

$Fe(OH)_2$ **may be dehydrated to iron oxide** FeO **or further oxidized to** $Fe(OH)_3$ **and then dehydrated to iron rust,** Fe_2O_3**.**

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